F26 AerE 3310: Flight Control Systems I

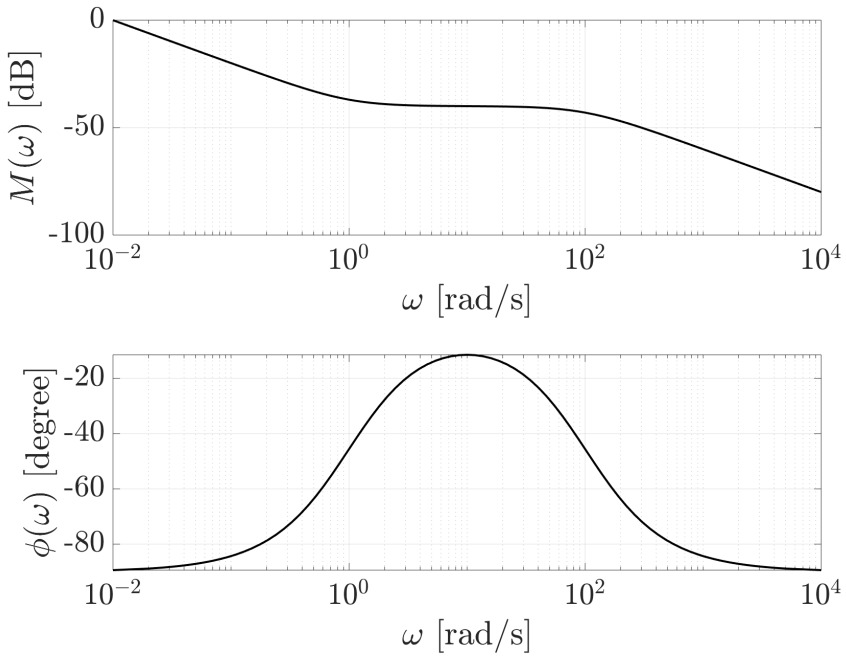
Practice Problem Set 9: Bode Analysis

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Problem 1

Use the following Bode plot of a SISO LTI system to answer both parts (a)-(b). Denote the transfer function for the given Bode plot as $G(s)$. In the Bode plot's vertical axes labels, the magnitude $M(\omega) = |G(j\omega)|$ and the phase $\phi(\omega) = \angle G(j\omega)$.



(a) Find the system's output $y(t)$ for the input $u(t) = 5\sin(0.01t)$. Show your calculations.

Solution for Problem 1(a)

From the phase plot, $\angle G(j0.01) = -90°$. From the magnitude plot,

$$20\log_{10}|G(j0.01)| = 0\,\text{dB} \quad \Rightarrow \quad |G(j0.01)| = 1.$$

So the output

$$y(t) = 5 \times |G(j0.01)|\sin\left(0.01t + \angle G(j0.01)\right) = 5\sin\left(0.01t - 90°\right).$$

(b) Find the system's output $y(t)$ for the input

$$u(t) = 5\sin(0.01t) - 7\sin(0.01t + 30^\circ) + 8\sin(0.01t - 20^\circ).$$

Solution for Problem 1(b)

Following the same argument as in part (a), the output for the input $7\sin(0.01t + 30^\circ)$ equals $7\sin(0.01t + 30^\circ - 90^\circ) = 7\sin(0.01t - 60^\circ)$. Similarly, the output for the input $8\sin(0.01t - 20^\circ)$ equals $8\sin(0.01t - 20^\circ - 90^\circ) = 8\sin(0.01t - 110^\circ)$. Since the system is linear, superposition principle gives the overall output

$$y(t) = 5\sin(0.01t - 90^\circ) - 7\sin(0.01t - 60^\circ) + 8\sin(0.01t - 110^\circ).$$

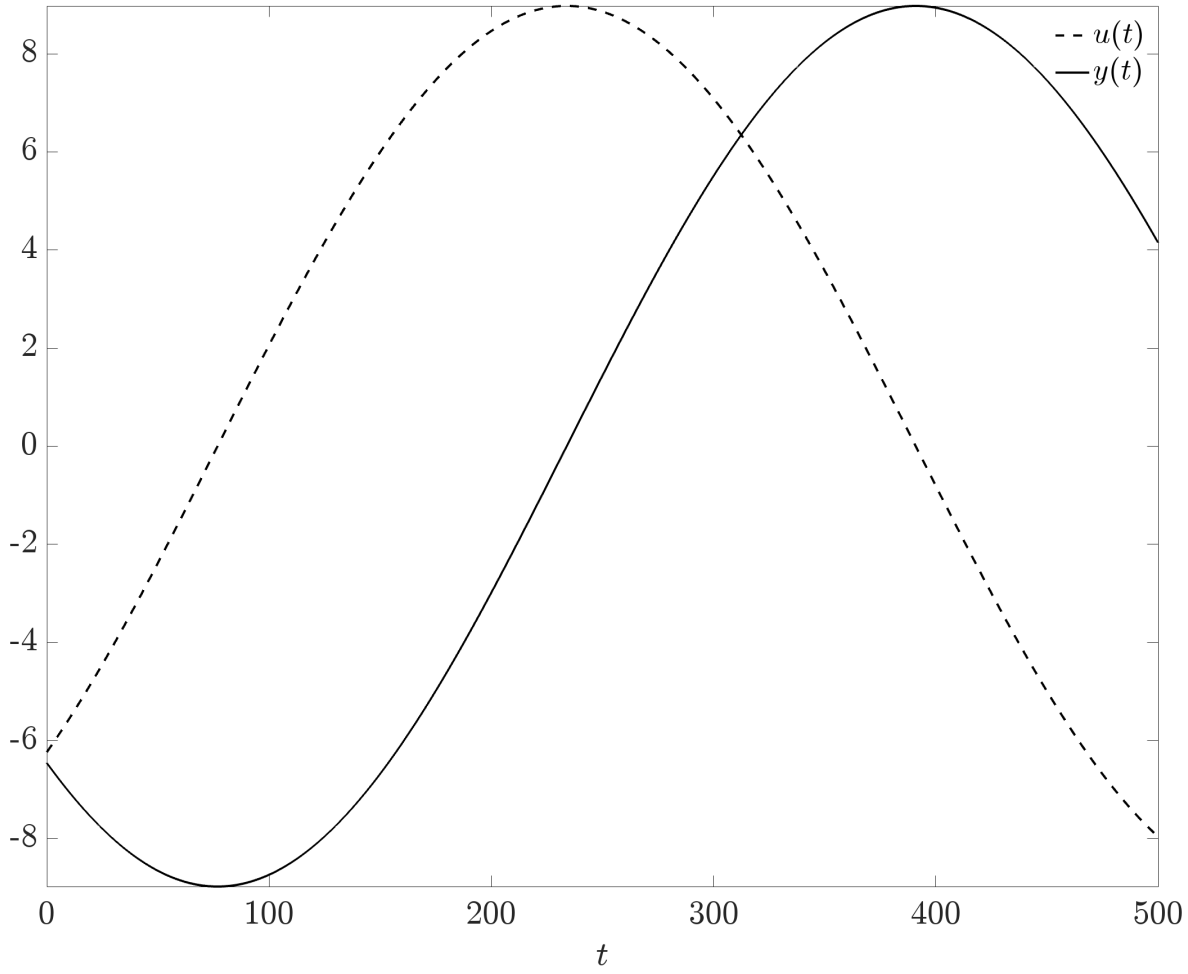


Figure 1: The input $u(t)$ and the output $y(t)$ for part (b).

Problem 2

Data through internet travel through Transmission Control Protocol (TCP) which can be modeled by the series composition of an integrator transfer function $\frac{k}{s}$ for some $k > 0$, and a time delay transfer function $e^{-s\tau}$ where $\tau \geq 0$. So the composed transfer function for the TCP is

$$G(s) = \frac{ke^{-s\tau}}{s}.$$

(a) Find the gain crossover frequency ω_{gc} for $G(s)$. Use this ω_{gc} to compute the phase margin.

Solution for Problem 2(a)

By definition of gain crossover frequency (Lec. 26 p. 6), we have

$$|G(j\omega_{\text{gc}})| = 1 \quad \Rightarrow \quad \frac{k}{|j\omega_{\text{gc}}|} = \frac{k}{\omega_{\text{gc}}} = 1 \quad \Rightarrow \quad \omega_{\text{gc}} = k.$$

By definition of phase margin (Lec. 26 p. 6), we get

$$\text{Phase margin} = \angle G(j\omega_{\text{gc}}) - (-\pi) = \angle k + \angle e^{-j\omega_{\text{gc}}\tau} - \angle j\omega_{\text{gc}} + \pi = 0 - \omega_{\text{gc}}\tau - \frac{\pi}{2} + \pi = \frac{\pi}{2} - k\tau.$$

(b) Use your answer in part (a) to compute the value of time delay τ which will cause zero phase margin. This value is called the delay margin.

Solution for Problem 2(b)

From part (a), the delay margin τ solves

$$0 = \frac{\pi}{2} - k\tau \quad \Rightarrow \quad \tau = \frac{\pi}{2k}.$$

(c) Find the phase crossover frequency ω_{pc} for $G(s)$. Use this ω_{pc} to compute the gain margin.

Solution for Problem 2(c)

By definition of phase crossover frequency (Lec. 26 p. 6), we have

$$\angle G(j\omega_{\text{pc}}) = -\pi \quad \Rightarrow \quad \angle k + \angle e^{-j\omega_{\text{pc}}\tau} - \angle j\omega_{\text{pc}} = -\pi \quad \Rightarrow \quad 0 - \omega_{\text{pc}}\tau - \frac{\pi}{2} = -\pi \quad \Rightarrow \quad \omega_{\text{pc}} = \frac{\pi}{2\tau}.$$

Applying the definition of the non-dimensional gain margin (Lec. 26 p. 7), we get

$$\frac{1}{|G(j\omega_{\text{pc}})|} = \frac{1}{k/\omega_{\text{pc}}} = \frac{\pi}{2k\tau}.$$

Equivalently, the gain margin in dB (Lec. 26 p. 7) equals

$$-20\log_{10}|G(j\omega_{\text{pc}})| = 20\log_{10}\left(\frac{\pi}{2k\tau}\right).$$